

# RET fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes RET gene fusions from the msk\_impact\_50k\_2026 study, covering 194 fusion events. 146/194 fusions (75.3%) were in-frame. 179/194 fusions (92.3%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ( $p=0.000419666$ , statistically significant at  $\alpha=0.05$ ). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 75.3% in-frame and 92.3% domain retention.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

### Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 141/146 (96.6%) of in-frame fusions retained the domain,  $p=0.000419666$  (statistically significant at  $\alpha=0.05$ ). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Domain disruption

Disruption of the PF00028 was tested with Fisher's exact test comparing in-frame fusions against all others; 139/146 (95.2%) of in-frame fusions disrupted the domain,  $p=0.00178046$  (statistically significant at  $\alpha=0.05$ ). The PF00028 appears to require loss or disruption rather than retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.663366.

### Cutpoint detection

Cutpoint detection scanned 194 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 627 aa (permutation-corrected  $p=0.00990099$ , statistically significant at  $\alpha=0.05$ ). This is -97 aa from the nearest configured domain boundary at 724 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=179$ ) and "not\_retained" ( $n=15$ ). For Frame\_status, retained: 141/157 (89.8%, 95% CI 84.1%-93.6%); not\_retained: 5/14 (35.7%, 95% CI 16.3%-61.2%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 67.4413 vs 49.2667,  $p=0.22222$  (not statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score produced results for this run (33 row(s) in composite\_evidence\_ranking); see the accompanying table.

## **Exon retention**

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## **Joint partner**

Joint partner reported: RET has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

### composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| KIF5B        | 87          | 0.4484536082474227   | 0.20043213737826426    | 0.017824657108181614    | 0.8093693978698724       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.38239793873644967 | 1    |
| CCDC6        | 51          | 0.26288659793814434  | 0.20043213737826426    | 0.017824657108181614    | 0.8167835941716136       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.3278399650889274  | 2    |
| PDCD10       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 1.0                      | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780028383343194  | 3    |
| TFG          | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 1.0                      | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2780028383343194  | 4    |
| ERC1         | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.9724770642201834       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.27387439796734686 | 5    |
| NCOA4        | 16          | 0.08247422680412371  | 0.20043213737826426    | 0.017824657108181614    | 0.8031823394495413       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.27167606554041035 | 6    |
| RBPM5        | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.9311926605504587       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.26768173741688817 | 7    |
| RUFY1        | 2           | 0.010309278350515464 | 0.20043213737826426    | 0.017824657108181614    | 0.9174311926605505       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.26716390898597925 | 8    |
| LIN52        | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.908256880733945        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2642413704444111  | 9    |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| EML4         | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.8944954128440367       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.26217715026092486 | 10   |
| DOCK1        | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.8371559633027523       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2535762328297322  | 11   |
| SPECC1L      | 4           | 0.020618556701030927 | 0.20043213737826426    | 0.017824657108181614    | 0.8044724770642202       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2533128851516844  | 12   |
| TIMM23B      | 3           | 0.015463917525773196 | 0.20043213737826426    | 0.017824657108181614    | 0.8035168195718654       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2516231447752538  | 13   |
| KCNMA1       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.823394495412844        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25151201264624595 | 14   |
| ERCC6        | 3           | 0.015463917525773196 | 0.20043213737826426    | 0.017824657108181614    | 0.8027522935779816       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25150846587617126 | 15   |
| RUFY3        | 2           | 0.010309278350515464 | 0.20043213737826426    | 0.017824657108181614    | 0.805045871559633        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.25030611082084164 | 16   |
| CCDC186      | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.805045871559633        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2487597190682643  | 17   |
| CUBN         | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.805045871559633        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2487597190682643  | 18   |
| GEMIN5       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.805045871559633        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2487597190682643  | 19   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| KIF13A       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.805045871559633        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2487597190682643  | 20   |
| RELCH        | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.805045871559633        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2487597190682643  | 21   |
| SLC4A4       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.805045871559633        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.2487597190682643  | 22   |
| CLIP1        | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.8027522935779816       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24841568237101658 | 23   |
| GRIPAP1      | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.8027522935779816       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24841568237101658 | 24   |
| RASSF4       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.8027522935779816       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24841568237101658 | 25   |
| RUFY2        | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.8027522935779816       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24841568237101658 | 26   |
| SHROOM3      | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.8027522935779816       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24841568237101658 | 27   |
| TRIM24       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.8027522935779816       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.24841568237101658 | 28   |
| KIAA1217     | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.536697247706422        | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.20850742549028264 | 29   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                                                 | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|---------------------|------|
| CTNNA3       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.29587155963302747      | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.17238357227927348 | 30   |
| CTNNA1       | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.1009174311926605       | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.14314045301321843 | 31   |
| UXS1         | 1           | 0.005154639175257732 | 0.20043213737826426    | 0.017824657108181614    | 0.06422018348623848      | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.13763586585725512 | 32   |
| FBXL7        | 2           | 0.010309278350515464 | 0.20043213737826426    | 0.017824657108181614    | 0.0                      | 0.7278348081553965         | ['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence'] | 0.12954923008689667 | 33   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value               | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|---------------------|-----------------------|---------------------|
| 219      | 0                      | 179              | 1                      | 14               | 0.0                 | 0.07731958762886598   | 1.1117104708745449  |
| 425      | 1                      | 178              | 1                      | 14               | 0.07865168539325842 | 0.14903050050745154   | 0.826724840000421   |
| 550      | 3                      | 176              | 2                      | 13               | 0.11079545454545454 | 0.048844025264081076  | 1.3111885527954676  |
| 556      | 4                      | 175              | 2                      | 13               | 0.14857142857142858 | 0.06998082470265705   | 1.1550209437950567  |
| 587      | 5                      | 174              | 3                      | 12               | 0.11494252873563218 | 0.016699929087495975  | 1.7772853729826312  |
| 622      | 6                      | 173              | 3                      | 12               | 0.13872832369942195 | 0.023865412628667555  | 1.6222310523597865  |
| 627      | 8                      | 171              | 5                      | 10               | 0.0935672514619883  | 0.001234261100135449  | 2.9085929583309174  |
| 639      | 9                      | 170              | 5                      | 10               | 0.10588235294117647 | 0.0018331517060542683 | 2.7368015926613842  |
| 640      | 10                     | 169              | 5                      | 10               | 0.11834319526627218 | 0.0026250181738802633 | 2.5808676854814183  |
| 657      | 11                     | 168              | 5                      | 10               | 0.13095238095238096 | 0.003644511061609513  | 2.438360727287861   |
| 663      | 13                     | 166              | 5                      | 10               | 0.1566265060240964  | 0.006510395780065948  | 2.1863926089934114  |
| 671      | 15                     | 164              | 5                      | 10               | 0.18292682926829268 | 0.01072245099455107   | 1.9697059299659876  |
| 673      | 16                     | 163              | 5                      | 10               | 0.19631901840490798 | 0.01342338349072885   | 1.8721380023380016  |
| 687      | 17                     | 162              | 5                      | 10               | 0.20987654320987653 | 0.01656697136370287   | 1.7807568784395744  |
| 708      | 18                     | 161              | 5                      | 10               | 0.2236024844720497  | 0.020185752752359003  | 1.694955050558838   |
| 712      | 88                     | 91               | 8                      | 7                | 0.8461538461538461  | 0.7940441928844832    | 0.10015532604674418 |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio        | p_value             | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|-------------------|---------------------|--------------------|
| 713      | 176                    | 3                | 14                     | 1                | 4.190476190476191 | 0.27712145920070164 | 0.5573298428514589 |
| 934      | 176                    | 3                | 15                     | 0                | 0.0               | 1.0                 | -0.0               |
| 1019     | 177                    | 2                | 15                     | 0                | 0.0               | 1.0                 | -0.0               |

## domain\_disruption tables

*frame\_domain\_contingency\_table*

|     |    |
|-----|----|
| 139 | 38 |
| 7   | 10 |

*domain\_disruption\_descriptives*

| Domain_key              | Domain_name | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-------------------------|-------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_disruption_pf00028 | PF00028     | 194                     | 17                   | 1               | 176              | 0.003161151466236212                            |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|     |    |
|-----|----|
| 141 | 38 |
| 5   | 10 |

*domain\_retention\_descriptives*

| Domain_key | Domain_name           | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|-----------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein kinase domain | 194                     | 179                  | 1               | 14               | 0.04988179669030733                             |

## exon\_retention tables

*Exon\_retention*

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 12   | 180                  | 194               | 0.9278350515463918 |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| CCDC186      | 1           |
| CCDC6        | 51          |
| CLIP1        | 1           |
| CTNNA1       | 1           |
| CTNNA3       | 1           |
| CUBN         | 1           |
| DOCK1        | 1           |
| EML4         | 1           |
| ERC1         | 1           |
| ERCC6        | 3           |
| FBXL7        | 2           |
| GEMIN5       | 1           |
| GRIPAP1      | 1           |
| KCNMA1       | 1           |
| KIAA1217     | 1           |
| KIF13A       | 1           |
| KIF5B        | 87          |
| LIN52        | 1           |
| NCOA4        | 16          |
| PDCD10       | 1           |
| RASSF4       | 1           |
| RBPMS        | 1           |
| RELCH        | 1           |
| RUFY1        | 2           |
| RUFY2        | 1           |
| RUFY3        | 2           |
| SHROOM3      | 1           |
| SLC4A4       | 1           |
| SPECC1L      | 4           |
| TFG          | 1           |
| TIMM23B      | 3           |
| TRIM24       | 1           |
| UXS1         | 1           |

## Per-event results (results.tsv)

| event_id                         | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                     | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-005<br>1781-T03-I<br>M6-3  | P-0051<br>781-T0<br>3-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threesome   | 43610377           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv   Protein Fusion: in frame {CCDC6:RET}     | NA                            |
| EVT-P-005<br>9857-T02-I<br>M7-4  | P-0059<br>857-T0<br>2-IM7 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threesome   | 43611615           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.453+8003:CCDC6_c.2137-417:RETinv   Protein Fusion: in frame {CCDC6:RET}      | NA                            |
| EVT-P-006<br>5992-T01-I<br>M7-5  | P-0065<br>992-T0<br>1-IM7 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threesome   | 43611698           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv   Protein Fusion: in frame {CCDC6:RET}     | NA                            |
| EVT-P-004<br>9461-T02-I<br>M6-6  | P-0049<br>461-T0<br>2-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threesome   | 43611705           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv   Protein Fusion: in frame {CCDC6:RET}      | NA                            |
| EVT-P-003<br>6379-T01-I<br>M6-7  | P-0036<br>379-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threesome   | 43611418           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv   Protein Fusion: in frame {CCDC6:RET}     | NA                            |
| EVT-P-003<br>7412-T01-I<br>M6-8  | P-0037<br>412-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threesome   | 43610839           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv   Protein Fusion: in frame {CCDC6:RET}     | NA                            |
| EVT-P-004<br>8611-T01-I<br>M6-9  | P-0048<br>611-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threesome   | 43611237           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv   Protein Fusion: in frame {CCDC6:RET}     | NA                            |
| EVT-P-006<br>4853-T01-I<br>M7-10 | P-0064<br>853-T0<br>1-IM7 |            | CCDC6-R<br>ET | CCDC6        | out-of-frame | threesome   | 43609511           | 10              | 627                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv   Protein Fusion: out of frame {CCDC6:RET} | NA                            |
| EVT-P-004<br>7144-T01-I<br>M6-11 | P-0047<br>144-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threesome   | 43611888           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8847:CCDC6_c.2137-144:RETinv   Protein Fusion: in frame {CCDC6:RET}      | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                         | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>8257-T01-I<br>M6-12 | P-0038<br>257-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threonine_prime | 43610594           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused with RET exon12);c.304-2131:CCDC6_c.2136+410:RETinv   Protein Fusion: in frame {CCDC6:RET}      | NA                            |
| EVT-P-001<br>7365-T04-I<br>M6-13 | P-0017<br>365-T0<br>4-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threonine_prime | 43611464           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                            |
| EVT-P-003<br>9548-T01-I<br>M6-14 | P-0039<br>548-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threonine_prime | 43610843           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv   Protein Fusion: in frame {CCDC6:RET}                                          | NA                            |
| EVT-P-004<br>0111-T01-I<br>M6-15 | P-0040<br>111-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | unknown      | threonine_prime | 43609967           | 11              | 640                         | False                | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv   Protein Fusion: mid-exon {CCDC6:RET}                                             | NA                            |
| EVT-P-004<br>1192-T01-I<br>M6-16 | P-0041<br>192-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threonine_prime | 43610571           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv   Protein Fusion: in frame {CCDC6:RET}                                          | NA                            |
| EVT-P-002<br>2642-T04-I<br>M6-17 | P-0022<br>642-T0<br>4-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threonine_prime | 43611768           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv   Protein Fusion: in frame {CCDC6:RET}                                         | NA                            |
| EVT-P-002<br>2642-T03-I<br>M6-18 | P-0022<br>642-T0<br>3-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threonine_prime | 43611768           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv   Protein Fusion: in frame {CCDC6:RET}                                         | NA                            |
| EVT-P-005<br>1781-T01-I<br>M6-19 | P-0051<br>781-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threonine_prime | 43610377           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv   Protein Fusion: in frame {CCDC6:RET}                                         | NA                            |
| EVT-P-004<br>2417-T01-I<br>M6-20 | P-0042<br>417-T0<br>1-IM6 |            | CCDC6-R<br>ET | CCDC6        | in-frame     | threonine_prime | 43610397           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein_kinase_domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv   Protein Fusion: in frame {CCDC6:RET}                                         | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name  | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|--------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0004901-T01-I M5-21 | P-0004901-T01-IM5 |            | CCDC6-RET    | CCDC6        | in-frame     | threonine_prime | 43611729           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_RETinv   Protein fusion: in frame (CCDC6-RET)                      | NA                           |
| EVT-P-0022380-T03-I M6-22 | P-0022380-T03-IM6 |            | CCDC6-RET    | CCDC6        | in-frame     | threonine_prime | 43610659           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv   Protein Fusion: in frame {CCDC6:RET}                                               | NA                           |
| EVT-P-0068500-T01-I M7-23 | P-0068500-T01-IM7 |            | CCDC6-RET    | CCDC6        | in-frame     | threonine_prime | 43610322           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv   Protein Fusion: in frame {CCDC6:RET}                                               | NA                           |
| EVT-P-0057735-T01-I M6-24 | P-0057735-T01-IM6 |            | CCDC6-RET    | CCDC6        | in-frame     | threonine_prime | 43611887           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RETinv   Protein Fusion: in frame {CCDC6:RET}                                                 | NA                           |
| EVT-P-0038053-T01-I M6-26 | P-0038053-T01-IM6 |            | CLIP1-RET    | CLIP1        | unknown      | threonine_prime | 43611121           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.21;q24.31)(chr10:g.43611121::chr12:g.122825512)   Protein Fusion: mid-exon {CLIP1:RET}            | NA                           |
| EVT-P-0000252-T01-I M3-27 | P-0000252-T01-IM3 |            | CUBN-RET     | CUBN         | unknown      | threonine_prime | 43610494           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | NA   Antisense fusion                                                                                                                                                | NA                           |
| EVT-P-0038256-T01-I M6-28 | P-0038256-T01-IM6 |            | ERC1-RET     | ERC1         | unknown      | threonine_prime | 43609964           | 11              | 639                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | ERC1 (NM_178040) - RET (NM_020975) Rearrangement : t(10;12)(q11.1;p13.32)(chr10:g.43609964::chr12:g.1264106)   Protein Fusion: mid-exon {ERC1:RET}                   | NA                           |
| EVT-P-0034781-T01-I M6-29 | P-0034781-T01-IM6 |            | GRIPAP1-RET  | GRIPAP1      | in-frame     | threonine_prime | 43611853           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | GRIPAP1 (NM_020137) - RET (NM_020975) rearrangement: t(X;10)(p11.23q11.21)(chrX:g.48831008::chr10:g.43611853)   Protein Fusion: in frame {GRIPAP1:RET}               | NA                           |
| EVT-P-0026245-T01-I M6-30 | P-0026245-T01-IM6 |            | KIAA1217-RET | KIAA1217     | unknown      | threonine_prime | 43606666           | 7               | 425                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIAA1217 (NM_019590) - RET (NM_020975) fusion (KIAA1217 exons 1-13 fused to RET exons 7-20): c.2390:KIAA1217_c.1275:RETdel   Protein Fusion: mid-exon {KIAA1217:RET} | NA                           |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusio_n_name       | part_ner_ge_ne | fra_me_st_atu_s | tar_get_ro_le               | breakp_oint_ge_nomic | break_point_exo_n | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains                  | lost_do_mains   | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                                                                                                                            | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|--------------|--------------------|----------------|-----------------|-----------------------------|----------------------|-------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|-------------------------------------|-----------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-004<br>2153-T01-I<br>M6-31 | P-0042<br>153-T0<br>1-IM6 |              | KIF1<br>3A-R<br>ET | KIF<br>13A     | in-fr<br>ame    | thr<br>ee<br>_pr<br>im<br>e | 436105<br>10         | 11                | 712                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF13A (NM_022113) - RET (NM_020975) fusion: t(6;10)(p22.3;q11.21)(chr6:g.17806558::chr10:g.43610510)   Protein Fusion: in frame {KIF13A:RET}                                                                                                                                                     | NA                            |
| EVT-P-001<br>9535-T02-I<br>M6-32 | P-0019<br>535-T0<br>2-IM6 |              | KIF5<br>B-RE<br>T  | KIF<br>5B      | in-fr<br>ame    | thr<br>ee<br>_pr<br>im<br>e | 436119<br>13         | 12                | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                                                                                | NA                            |
| EVT-P-003<br>9257-T03-I<br>M7-33 | P-0039<br>257-T0<br>3-IM7 |              | KIF5<br>B-RE<br>T  | KIF<br>5B      | unk<br>now<br>n | thr<br>ee<br>_pr<br>im<br>e | 436100<br>60         | 11                | 671                          | False                    | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv   Protein Fusion: mid-exon {KIF5B:RET}                                                                                                                                                                                | NA                            |
| EVT-P-000<br>9514-T01-I<br>M5-34 | P-0009<br>514-T0<br>1-IM5 |              | KIF5<br>B-RE<br>T  | KIF<br>5B      | in-fr<br>ame    | thr<br>ee<br>_pr<br>im<br>e | 436118<br>11         | 12                | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozantinib. PMID: 23533264   Protein fusion: in frame (KIF5B-RET) | NA                            |
| EVT-P-005<br>5183-T01-I<br>M6-35 | P-0055<br>183-T0<br>1-IM6 |              | KIF5<br>B-RE<br>T  | KIF<br>5B      | in-fr<br>ame    | thr<br>ee<br>_pr<br>im<br>e | 436103<br>08         | 11                | 712                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                                                                                                                            | NA                            |
| EVT-P-001<br>1035-T01-I<br>M5-36 | P-0011<br>035-T0<br>1-IM5 |              | KIF5<br>B-RE<br>T  | KIF<br>5B      | in-fr<br>ame    | thr<br>ee<br>_pr<br>im<br>e | 436113<br>22         | 12                | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1726-2296:KIF5B_c.2137-710:RETinv   Protein fusion: in frame (KIF5B-RET)                                                                                                                             | NA                            |
| EVT-P-001<br>2648-T01-I<br>M5-37 | P-0012<br>648-T0<br>1-IM5 |              | KIF5<br>B-RE<br>T  | KIF<br>5B      | in-fr<br>ame    | thr<br>ee<br>_pr<br>im<br>e | 436116<br>97         | 12                | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv   Protein fusion: in frame (KIF5B-RET)                                                                                                                                                                            | NA                            |
| EVT-P-005<br>3175-T01-I<br>M6-38 | P-0053<br>175-T0<br>1-IM6 |              | KIF5<br>B-RE<br>T  | KIF<br>5B      | in-fr<br>ame    | thr<br>ee<br>_pr<br>im<br>e | 436111<br>62         | 12                | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-282:KIF5B_c.2137-870:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                                                                                                                             | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                             | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-005<br>2861-T02-I<br>M6-39 | P-0052<br>861-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43610260           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-20:KIF5B_c.2136+76:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                | NA                           |
| EVT-P-001<br>2825-T01-I<br>M5-40 | P-0013<br>825-T0<br>1-IM5 |            | KIF5<br>B-RET | KIF5B        | unknown      | threesome   | 43611520           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1782:KIF5B_c.2137-512:RETinv   Protein fusion: mid-exon (KIF5B-RET)      | NA                           |
| EVT-P-005<br>2826-T01-I<br>M6-41 | P-0052<br>826-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611182           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2358:KIF5B_c.2137-850:RETinv   Protein Fusion: in frame {KIF5B:RET}                                             | NA                           |
| EVT-P-001<br>4808-T01-I<br>M6-42 | P-0014<br>808-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611055           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv   Protein fusion: in frame (KIF5B-RET) | NA                           |
| EVT-P-003<br>4357-T01-I<br>M6-43 | P-0034<br>357-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611187           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv   Protein Fusion: in frame {KIF5B:RET}                                             | NA                           |
| EVT-P-001<br>4808-T02-I<br>M6-44 | P-0014<br>808-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611056           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-003<br>3700-T02-I<br>M6-45 | P-0033<br>700-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611670           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET   Protein Fusion: in frame {KIF5B:RET}                                                | NA                           |
| EVT-P-001<br>4808-T03-I<br>M6-46 | P-0014<br>808-T0<br>3-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611056           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv   Protein Fusion: in frame {KIF5B:RET}                                             | NA                           |
| EVT-P-000<br>8289-T01-I<br>M5-47 | P-0008<br>289-T0<br>1-IM5 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611748           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv   Protein fusion: in frame (KIF5B-RET) | NA                           |

| event_id                         | sampl_e_id                | pa_tie_n_t_id | fusio_n_name      | part_ner_ge_ne | fra_me_st_atu_s | tar_get_ro_le         | breakp_o_int_ge_nomic | break_p_o_int_exo_n | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains                  | lost_do_mains   | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                      | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|---------------|-------------------|----------------|-----------------|-----------------------|-----------------------|---------------------|------------------------------|--------------------------|-----------------|----------------------------|----------------------|-------------------------------------|-----------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-000<br>0252-T01-I<br>M3-48 | P-0000<br>252-T0<br>1-IM3 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436105<br>97          | 11                  | 712                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | RET (NM_020975) Inversion (c.2136+413_KIF5B:c.1726-153inv)   Protein fusion: in frame (KIF5B-RET)                                                                           | NA                            |
| EVT-P-000<br>8101-T01-I<br>M5-49 | P-0008<br>101-T0<br>1-IM5 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436104<br>55          | 11                  | 712                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv   Protein fusion: in frame (KIF5B-RET)         | NA                            |
| EVT-P-004<br>8968-T01-I<br>M6-50 | P-0048<br>968-T0<br>1-IM6 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436118<br>09          | 12                  | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                       | NA                            |
| EVT-P-004<br>8829-T01-I<br>M6-51 | P-0048<br>829-T0<br>1-IM6 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436106<br>16          | 11                  | 712                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                       | NA                            |
| EVT-P-006<br>5876-T01-I<br>M7-52 | P-0065<br>876-T0<br>1-IM7 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436112<br>99          | 12                  | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                      | NA                            |
| EVT-P-001<br>6221-T01-I<br>M6-53 | P-0016<br>221-T0<br>1-IM6 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436112<br>68          | 12                  | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |
| EVT-P-003<br>7367-T01-I<br>M6-54 | P-0037<br>367-T0<br>1-IM6 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436118<br>74          | 12                  | 713                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                      | NA                            |
| EVT-P-005<br>7072-T02-I<br>M6-55 | P-0057<br>072-T0<br>2-IM6 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436111<br>06          | 11                  | 712                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                      | NA                            |
| EVT-P-001<br>7214-T01-I<br>M6-56 | P-0017<br>214-T0<br>1-IM6 |               | KIF5<br>B-RE<br>T | KIF<br>5B      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436106<br>47          | 11                  | 712                          | True                     | retain<br>ed    | 1.0                        | False                | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8 |                     | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused in-frame with RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                              | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>7214-T02-I<br>M6-57 | P-0017<br>214-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610647           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused to RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv   Protein Fusion: in frame {KIF5B:RET}    | NA                            |
| EVT-P-003<br>2753-T01-I<br>M6-58 | P-0032<br>753-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610925           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1323:KIF5B_c.2136+741:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                            |
| EVT-P-005<br>7072-T03-I<br>M7-59 | P-0057<br>072-T0<br>3-IM7 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611106           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                            |
| EVT-P-003<br>2686-T01-I<br>M6-60 | P-0032<br>686-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611982           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2299:KIF5B_c.2137-50:RETinv   Protein Fusion: in frame {KIF5B:RET}                                               | NA                            |
| EVT-P-001<br>7523-T02-I<br>M6-61 | P-0017<br>523-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611192           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion ( KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2362:KIF5B_c.2137-840:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |
| EVT-P-004<br>6827-T02-I<br>M6-62 | P-0046<br>827-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610889           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305:KIF5B_c.2136+705:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                            |
| EVT-P-003<br>2407-T01-I<br>M6-63 | P-0032<br>407-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611730           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230:KIF5B_c.2137-302:RETinv   Protein Fusion: in frame {KIF5B:RET}                                               | NA                            |
| EVT-P-001<br>8069-T01-I<br>M6-64 | P-0018<br>069-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | out-of-frame | five_prime      | 43607832           | 8               | 550                         | True                 | lost          | 0.0                      | False               | Protein kinase domain | PF00028      |                   | RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv   Protein Fusion: out of frame {RET:KIF5B}                                     | NA                            |
| EVT-P-000<br>5573-T01-I<br>M5-65 | P-0005<br>573-T0<br>1-IM5 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610418           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1726-860:KIF5B_c.2136+234:RETinv   Protein fusion: in frame (KIF5B-RET)          | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                             | source_site_2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0004995-T01-I M5-66 | P-0004995-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | threonine_prime | 43610660           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B(NM_004521) -RET (NM_020975) Fusion ( KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+476:RETinv   Protein fusion: in frame (KIF5B-RET)        | NA                            |
| EVT-P-0036092-T01-I M6-67 | P-0036092-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | threonine_prime | 43610413           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |
| EVT-P-0046726-T01-I M6-68 | P-0046726-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | threonine_prime | 43610987           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                            |
| EVT-P-0032227-T01-I M6-69 | P-0032227-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | threonine_prime | 43611125           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv   Protein Fusion: in frame {KIF5B:RET}                                             | NA                            |
| EVT-P-0020273-T02-I M6-70 | P-0020273-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | threonine_prime | 43611819           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv   Protein Fusion: in frame {KIF5B:RET}  | NA                            |
| EVT-P-0043007-T02-I M6-71 | P-0043007-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | threonine_prime | 43610813           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                            |
| EVT-P-0021150-T01-I M6-72 | P-0021150-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | threonine_prime | 43611591           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv   Protein Fusion: in frame {KIF5B:RET}  | NA                            |
| EVT-P-0021578-T01-I M6-73 | P-0021578-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | threonine_prime | 43610809           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                            | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>1879-T01-I<br>M6-74 | P-0021<br>879-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610699           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B_c.2136+515_RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |
| EVT-P-002<br>1937-T01-I<br>M6-75 | P-0021<br>937-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611555           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:KIF5B_c.2137-477:RETinv   Protein Fusion: in frame {KIF5B:RET}                              | NA                            |
| EVT-P-002<br>1937-T03-I<br>M6-76 | P-0021<br>937-T0<br>3-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611555           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:KIF5B_c.2137-477:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                             | NA                            |
| EVT-P-004<br>3007-T01-I<br>M6-77 | P-0043<br>007-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610813           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                             | NA                            |
| EVT-P-004<br>2608-T01-I<br>M6-78 | P-0042<br>608-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611802           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1670:KIF5B_c.2137-230:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                            | NA                            |
| EVT-P-004<br>2580-T01-I<br>M6-79 | P-0042<br>580-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610944           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1619:KIF5B_c.2136+760:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                            | NA                            |
| EVT-P-005<br>9913-T01-I<br>M7-80 | P-0059<br>913-T0<br>1-IM7 |            | KIF5<br>B-RET | KIF5B        | unknown      | threonine_prime | 43610108           | 11              | 687                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2441:KIF5B_c.2060:RETinv   Protein Fusion: mid-exon {KIF5B:RET}                                                                                | NA                            |
| EVT-P-000<br>1010-T02-I<br>M5-81 | P-0001<br>010-T0<br>2-IM5 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610863           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-780:KIF5B_c.2136+679:RETinv   Protein fusion: in frame (KIF5B-RET)                                                                             | NA                            |
| EVT-P-006<br>0748-T02-I<br>M7-82 | P-0060<br>748-T0<br>2-IM7 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611527           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                            | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                            | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>2104-T01-I<br>M6-83 | P-0042<br>104-T0<br>1-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610505           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1056:KIF5B_c.2136+321:RETinv   Protein Fusion: in frame {KIF5B:RET}                                            | NA                           |
| EVT-P-000<br>3125-T01-I<br>M5-84 | P-0003<br>125-T0<br>1-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611584           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NA   Protein fusion: in frame (KIF5B-RET)                                                                                                                         | NA                           |
| EVT-P-000<br>3125-T01-I<br>M5-85 | P-0003<br>125-T0<br>1-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | five_prime  | 43611581           | 12              | 713                         | True                   | lost          | 0.0                      | False               | PF00028               | Protein kinase domain |                   | NA   Protein fusion: in frame (RET-KIF5B)                                                                                                                         | NA                           |
| EVT-P-000<br>2873-T01-I<br>M3-86 | P-0002<br>873-T0<br>1-IM3 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611244           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NA   Protein fusion: in frame (KIF5B-RET)                                                                                                                         | NA                           |
| EVT-P-002<br>2866-T01-I<br>M6-87 | P-0022<br>866-T0<br>1-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611871           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_c.2137-161:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-002<br>4225-T01-I<br>M6-88 | P-0024<br>225-T0<br>1-IM6 |            | KIF5B-RET   | KIF5B        | unknown      | three_prime | 43609110           | 10              | 622                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_c.1866:RETinv   Protein Fusion: mid-exon {KIF5B:RET}     | NA                           |
| EVT-P-000<br>2694-T02-I<br>M5-89 | P-0002<br>694-T0<br>2-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611682           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NA   Protein fusion: in frame (KIF5B-RET)                                                                                                                         | NA                           |
| EVT-P-002<br>4843-T01-I<br>M6-90 | P-0024<br>843-T0<br>1-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611733           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_c.2137-299:RETinv   Protein Fusion: in frame {KIF5B:RET}    | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>5100-T01-I<br>M6-91 | P-0025<br>100-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611511           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2273:KIF5B_c.2137-521:RETinv   Protein Fusion: in frame {KIF5B:RET}    | NA                            |
| EVT-P-002<br>5100-T02-I<br>M6-92 | P-0025<br>100-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611511           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) and RET (NM_020975) Fusion (KIF5B exon 16 fused to RET exon 12): c.1726-2273:KIF5B_c.2137-788:RETinv   Protein Fusion: in frame {KIF5B:RET}         | NA                            |
| EVT-P-003<br>9941-T01-I<br>M6-93 | P-0039<br>941-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611339           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                | NA                            |
| EVT-P-006<br>0748-T03-I<br>M7-94 | P-0060<br>748-T0<br>3-IM7 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611527           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                | NA                            |
| EVT-P-003<br>9474-T02-I<br>M6-95 | P-0039<br>474-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43610359           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                 | NA                            |
| EVT-P-003<br>9474-T01-I<br>M6-96 | P-0039<br>474-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43610359           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                 | NA                            |
| EVT-P-005<br>5183-T02-I<br>M6-97 | P-0055<br>183-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43610308           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                | NA                            |
| EVT-P-002<br>5838-T01-I<br>M6-98 | P-0025<br>838-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43610972           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion ( KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |
| EVT-P-002<br>5928-T01-I<br>M6-99 | P-0025<br>928-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threesome   | 43611910           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv   Protein Fusion: in frame {KIF5B:RET}  | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                             | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>5928-T02-I<br>M6-100 | P-0025<br>928-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | three_prime | 43611910           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                            |
| EVT-P-003<br>6637-T01-I<br>M6-101 | P-0036<br>637-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | three_prime | 43611965           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv   Protein Fusion: in frame {KIF5B:RET}  | NA                            |
| EVT-P-003<br>9257-T01-I<br>M6-102 | P-0039<br>257-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | unknown      | three_prime | 43610060           | 11              | 671                         | False                | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv   Protein Fusion: mid-exon {KIF5B:RET}                                                 | NA                            |
| EVT-P-002<br>6693-T01-I<br>M6-103 | P-0026<br>693-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | three_prime | 43610511           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+156:KIF5_c.2136+327:RETinv.   Protein Fusion: in frame {KIF5B:RET}                                              | NA                            |
| EVT-P-002<br>6777-T01-I<br>M6-104 | P-0026<br>777-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | three_prime | 43610717           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |
| EVT-P-002<br>6777-T02-I<br>M6-105 | P-0026<br>777-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | three_prime | 43610717           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |
| EVT-P-003<br>8961-T01-I<br>M6-106 | P-0038<br>961-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | three_prime | 43610775           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                            |
| EVT-P-002<br>7723-T01-I<br>M6-107 | P-0027<br>723-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | unknown      | three_prime | 43610171           | 11              | 708                         | False                | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20): c.1725+966:KIF5B_c.2123:RETinv   Protein Fusion: mid-exon {KIF5B:RET}      | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name   | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|---------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>6737-T02-I<br>M6-108 | P-0036<br>737-T0<br>2-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43612025           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:KIF5B_c.2137-7:RETinv   Protein Fusion: in frame {KIF5B:RET}      | NA                            |
| EVT-P-006<br>1331-T01-I<br>M7-109 | P-0061<br>331-T0<br>1-IM7 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610884           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                 | NA                            |
| EVT-P-002<br>7991-T01-I<br>M6-110 | P-0027<br>991-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43608201           | 9               | 550                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv   Protein Fusion: in frame {KIF5B:RET}      | NA                            |
| EVT-P-006<br>2948-T03-I<br>M7-111 | P-0062<br>948-T0<br>3-IM7 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610211           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                 | NA                            |
| EVT-P-006<br>4951-T01-I<br>M7-112 | P-0064<br>951-T0<br>1-IM7 |            | KIF5<br>B-RET | KIF5B        | out-of-frame | threonine_prime | 43611130           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv   Protein Fusion: out of frame {KIF5B:RET}                                               | NA                            |
| EVT-P-003<br>0453-T01-I<br>M6-113 | P-0030<br>453-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43610911           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                | NA                            |
| EVT-P-003<br>7494-T01-I<br>M6-114 | P-0037<br>494-T0<br>1-IM6 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611755           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12 ): c.1725+985:KIF5B_c.2137-277:RETinv   Protein Fusion: in frame {KIF5B:RET}           | NA                            |
| EVT-P-000<br>0208-T02-I<br>M5-115 | P-0000<br>208-T0<br>2-IM5 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611647           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv   Protein fusion: in frame (KIF5B-RET) | NA                            |
| EVT-P-000<br>0208-T01-I<br>M3-116 | P-0000<br>208-T0<br>1-IM3 |            | KIF5<br>B-RET | KIF5B        | in-frame     | threonine_prime | 43611647           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | RET (NM_020975) Inversion ( c.2137-385_KIF5B:c.1726-1971inv)   Protein fusion: in frame (KIF5B-RET)                                                                   | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                                    | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>6221-T02-I<br>M6-117 | P-0016<br>221-T0<br>2-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | threesome   | 43611268           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                            |
| EVT-P-001<br>9663-T01-I<br>M6-118 | P-0019<br>663-T0<br>1-IM6 |            | LIN52-RET   | LIN52        | out-of-frame | five_prime  | 43608613           | 9               | 587                         | True                 | lost          | 0.0                      | False               | PF00028               | Protein kinase domain |                   | RET (NM_020975) - LIN52 (NM_001024674) Rearrangement : t(10;14)(p32.1;q12)(chr10:g.43608613:chr14:g.74562750)   Protein Fusion: out of frame {RET:LIN52}                                  | NA                            |
| EVT-P-006<br>5826-T01-I<br>M7-120 | P-0065<br>826-T0<br>1-IM7 |            | NCOA4-RET   | NCOA4        | out-of-frame | threesome   | 43610857           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2136+673:RET_c.*23+937NCOA4dup   Protein Fusion: out of frame {TIMM23B:RET}                                                              | NA                            |
| EVT-P-003<br>7961-T01-I<br>M6-121 | P-0037<br>961-T0<br>1-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | threesome   | 43611448           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.763-360:NCOA4_c.2137-584:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                      | NA                            |
| EVT-P-003<br>8315-T01-I<br>M6-122 | P-0038<br>315-T0<br>1-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | threesome   | 43611981           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                              | NA                            |
| EVT-P-006<br>0765-T02-I<br>M7-123 | P-0060<br>765-T0<br>2-IM7 |            | NCOA4-RET   | NCOA4        | out-of-frame | threesome   | 43611844           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup   Protein Fusion: out of frame {TIMM23B:RET}                                                              | NA                            |
| EVT-P-004<br>3077-T01-I<br>M6-124 | P-0043<br>077-T0<br>1-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | threesome   | 43611580           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                             | NA                            |
| EVT-P-000<br>5748-T01-I<br>M5-125 | P-0005<br>748-T0<br>1-IM5 |            | NCOA4-RET   | NCOA4        | in-frame     | threesome   | 43611325           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.763-260:NCOA4_c.2137-707:RETdup   Protein fusion: in frame (NCOA4-RET)                                                                  | NA                            |
| EVT-P-004<br>7991-T02-I<br>M6-126 | P-0047<br>991-T0<br>2-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | threesome   | 43611538           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+537:NCOA4_c.2137-494:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                             | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>8805-T01-I<br>M6-127 | P-0048<br>805-T0<br>1-IM6 |            | NCOA4-R<br>ET   | NCOA4        | unknown      | threesome   | 43611225           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup   Protein Fusion: mid-exon {NCOA4:RET}                                                                                   | NA                           |
| EVT-P-004<br>9600-T01-I<br>M6-128 | P-0049<br>600-T0<br>1-IM6 |            | NCOA4-R<br>ET   | NCOA4        | unknown      | threesome   | 43611613           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup   Protein Fusion: mid-exon {NCOA4:RET}                                                                                  | NA                           |
| EVT-P-005<br>6870-T01-I<br>M6-129 | P-0056<br>870-T0<br>1-IM6 |            | NCOA4-R<br>ET   | NCOA4        | out-of-frame | threesome   | 43610337           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                                         | NA                           |
| EVT-P-005<br>2409-T01-I<br>M6-130 | P-0052<br>409-T0<br>1-IM6 |            | NCOA4-R<br>ET   | NCOA4        | out-of-frame | threesome   | 43611691           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                                  | NA                           |
| EVT-P-000<br>4094-T02-I<br>M6-131 | P-0004<br>094-T0<br>2-IM6 |            | RELCH-R<br>ET   | RELCH        | in-frame     | threesome   | 43610279           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | KIAA1468 (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.21;q21.33)(chr10:g.43610279::chr18:g.59909399)   Protein Fusion: in frame {KIAA1468:RET} | NA                           |
| EVT-P-001<br>5588-T01-I<br>M6-132 | P-0015<br>588-T0<br>1-IM6 |            | RET-CCD<br>C186 | CCDC186      | in-frame     | threesome   | 43610308           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | C10orf118 (NM_018017) - RET (NM_020975) Rearrangement : c.1326+142:C10orf118_c.2136+124:RETinv   Protein fusion: in frame (C10orf118-RET)                                                             | NA                           |
| EVT-P-002<br>0974-T01-I<br>M6-133 | P-0020<br>974-T0<br>1-IM6 |            | RET-CCD<br>C6   | CCDC6        | in-frame     | threesome   | 43611184           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:CCDC6_c.2137-848:RETinv   Protein Fusion: in frame {CCDC6:RET}                                             | NA                           |
| EVT-P-000<br>1745-T02-I<br>M5-134 | P-0001<br>745-T0<br>2-IM5 |            | RET-CCD<br>C6   | CCDC6        | in-frame     | threesome   | 43611245           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-787:RETinv   Protein fusion: in frame (CCDC6-RET)                                                                         | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                         | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|-----------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>7365-T02-I<br>M6-135 | P-0017<br>365-T0<br>2-IM6 |            | RET-<br>CCD<br>C6 | CC<br>DC6    | in-frame     | three_prime | 436114<br>64       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                            |
| EVT-P-000<br>1898-T01-I<br>M3-136 | P-0001<br>898-T0<br>1-IM3 |            | RET-<br>CCD<br>C6 | CC<br>DC6    | in-frame     | three_prime | 436116<br>99       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5   Protein fusion: in frame (CCDC6-RET)                                               | NA                            |
| EVT-P-002<br>9602-T01-I<br>M6-137 | P-0029<br>602-T0<br>1-IM6 |            | RET-<br>CCD<br>C6 | CC<br>DC6    | in-frame     | three_prime | 436111<br>60       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv   Protein Fusion: in frame {CCDC6:RET}                                          | NA                            |
| EVT-P-000<br>4916-T01-I<br>M5-138 | P-0004<br>916-T0<br>1-IM5 |            | RET-<br>CCD<br>C6 | CC<br>DC6    | in-frame     | three_prime | 436103<br>53       | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | CCDC6(NM_005436)-RET(NM_020975) Fusion(CCDC6 exon 2 fused to RET exon11) : c.304-1153:CCDC6_c.2136+169:RETinv   Protein fusion: in frame (CCDC6-RET)           | NA                            |
| EVT-P-000<br>0425-T01-I<br>M3-139 | P-0000<br>425-T0<br>1-IM3 |            | RET-<br>CCD<br>C6 | CC<br>DC6    | in-frame     | five_prime  | 436104<br>50       | 11              | 712                         | True                   | lost          | 0.0                      | False               | PF00<br>28                  | Protein<br>kinase<br>domain |                   | RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv)   Protein fusion: in frame (RET-CCDC6)                                               | NA                            |
| EVT-P-003<br>1330-T01-I<br>M6-140 | P-0031<br>330-T0<br>1-IM6 |            | RET-<br>CCD<br>C6 | CC<br>DC6    | in-frame     | three_prime | 436103<br>53       | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+3703:CCDC6_c.2136+169:RETinv   Protein Fusion: in frame {CCDC6:RET}  | NA                            |
| EVT-P-003<br>0990-T03-I<br>M6-141 | P-0030<br>990-T0<br>3-IM6 |            | RET-<br>CCD<br>C6 | CC<br>DC6    | in-frame     | three_prime | 436118<br>81       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-151:RETinv   Protein Fusion: in frame {CCDC6:RET}  | NA                            |
| EVT-P-000<br>5168-T01-I<br>M5-142 | P-0005<br>168-T0<br>1-IM5 |            | RET-<br>CCD<br>C6 | CC<br>DC6    | out-of-frame | three_prime | 436087<br>97       | 10              | 587                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 Exon1 fused to RET Exon10) : c.304-11493:CCDC6_c.1760-207:RETinv   Protein fusion: out of frame (CCDC6-RET)  | NA                            |

| event_id                          | sampl_e_id                | pa_tie_n_t_id | fusio_n_name      | part_ner_ge_ne | fra_me_st_atu_s | tar_get_ro_le         | breakp_o_int_ge_nomic | break_p_o_int_exo_n | breakpoint_protein_p_osition | is_intron_ic_break_p_o_int | dom_ain_statu_s | domain_r_etained_fr_action | domain_is_tru_n_cated | retain_ed_do_mains                  | lost_do_mains                            | disrupt_ed_do_mains | source_annotation_text                                                                                                                                               | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|---------------|-------------------|----------------|-----------------|-----------------------|-----------------------|---------------------|------------------------------|----------------------------|-----------------|----------------------------|-----------------------|-------------------------------------|------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-002<br>2380-T01-I<br>M6-143 | P-0022<br>380-T0<br>1-IM6 |               | RET-<br>CCD<br>C6 | CC<br>DC6      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436106<br>59          | 11                  | 712                          | True                       | retain<br>ed    | 1.0                        | False                 | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8                          |                     | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20 ): c.303+21810:CCDC6_c.2136+475:RETinv   Protein Fusion: in frame {CCDC6:RET}      | NA                            |
| EVT-P-002<br>2380-T02-I<br>M6-144 | P-0022<br>380-T0<br>2-IM6 |               | RET-<br>CCD<br>C6 | CC<br>DC6      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436106<br>59          | 11                  | 712                          | True                       | retain<br>ed    | 1.0                        | False                 | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8                          |                     | CCDC6 (NM_005436) - RET(NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:c.2136+475inv   Protein Fusion: in frame {CCDC6:RET} | NA                            |
| EVT-P-002<br>9315-T02-I<br>M6-145 | P-0029<br>315-T0<br>2-IM6 |               | RET-<br>CCD<br>C6 | CC<br>DC6      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436106<br>31          | 11                  | 712                          | True                       | retain<br>ed    | 1.0                        | False                 | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8                          |                     | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv   Protein Fusion: in frame {CCDC6:RET}       | NA                            |
| EVT-P-002<br>9315-T01-I<br>M6-146 | P-0029<br>315-T0<br>1-IM6 |               | RET-<br>CCD<br>C6 | CC<br>DC6      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436106<br>31          | 11                  | 712                          | True                       | retain<br>ed    | 1.0                        | False                 | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8                          |                     | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20: c.304-20462:CCDC6_c.2136+447:RETinv   Protein Fusion: in frame {CCDC6:RET}        | NA                            |
| EVT-P-000<br>0889-T01-I<br>M3-147 | P-0000<br>889-T0<br>1-IM3 |               | RET-<br>CCD<br>C6 | CC<br>DC6      | in-fr<br>ame    | five<br>_pr<br>im_e   | 436119<br>94          | 12                  | 713                          | True                       | lost            | 0.0                        | False                 | PF000<br>28                         | Prot<br>ein<br>kina<br>se d<br>oma<br>in |                     | RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv   Protein fusion: in frame (RET-CCDC6)                                              | NA                            |
| EVT-P-000<br>7322-T02-I<br>M6-148 | P-0007<br>322-T0<br>2-IM6 |               | RET-<br>CCD<br>C6 | CC<br>DC6      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436102<br>68          | 11                  | 712                          | True                       | retain<br>ed    | 1.0                        | False                 | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8                          |                     | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv   Protein Fusion: in frame {CCDC6:RET}                                                 | NA                            |
| EVT-P-002<br>7953-T01-I<br>M6-149 | P-0027<br>953-T0<br>1-IM6 |               | RET-<br>CCD<br>C6 | CC<br>DC6      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436112<br>78          | 12                  | 713                          | True                       | retain<br>ed    | 1.0                        | False                 | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8                          |                     | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv   Protein Fusion: in frame {CCDC6:RET}       | NA                            |
| EVT-P-000<br>8504-T03-I<br>M6-150 | P-0008<br>504-T0<br>3-IM6 |               | RET-<br>CCD<br>C6 | CC<br>DC6      | in-fr<br>ame    | thr_ee<br>_pr<br>im_e | 436105<br>83          | 11                  | 712                          | True                       | retain<br>ed    | 1.0                        | False                 | Protei<br>n<br>kinase<br>domai<br>n | PF0<br>002<br>8                          |                     | CCDC6 (NM_005436) - RET (NM_020975) fusion ( CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv   Protein Fusion: in frame {CCDC6:RET}    | NA                            |

| event_id                   | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intron_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                         | source_site_2_effect_on_frame |
|----------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|----------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0008955-T03-I M5-151 | P-0008955-T03-IM5 |            | RET-CCDC6   | CCDC6        | in-frame     | three_prime | 43611019           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv   Protein fusion: in frame (CCDC6-RET)                                         | NA                            |
| EVT-P-0013023-T01-I M5-152 | P-0013023-T01-IM5 |            | RET-CCDC6   | CCDC6        | in-frame     | three_prime | 43611810           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv   Protein fusion: in frame (CCDC6-RET)                                         | NA                            |
| EVT-P-0013394-T01-I M5-153 | P-0013394-T01-IM5 |            | RET-CCDC6   | CCDC6        | in-frame     | three_prime | 43610232           | 11              | 712                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv   Protein fusion: in frame (CCDC6-RET)                                          | NA                            |
| EVT-P-0022607-T01-I M6-154 | P-0022607-T01-IM6 |            | RET-CCDC6   | CCDC6        | in-frame     | three_prime | 43611212           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12 - 20): c.304-414:CCDC6_c.2137-820:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                            |
| EVT-P-0022642-T01-I M6-155 | P-0022642-T01-IM6 |            | RET-CCDC6   | CCDC6        | in-frame     | three_prime | 43611768           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv   Protein Fusion: in frame {CCDC6:RET}        | NA                            |
| EVT-P-0022642-T02-I M6-156 | P-0022642-T02-IM6 |            | RET-CCDC6   | CCDC6        | in-frame     | three_prime | 43611768           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                            |
| EVT-P-0017365-T03-I M6-157 | P-0017365-T03-IM6 |            | RET-CCDC6   | CCDC6        | in-frame     | three_prime | 43611464           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                            |
| EVT-P-0025349-T03-I M6-158 | P-0025349-T03-IM6 |            | RET-CCDC6   | CCDC6        | in-frame     | three_prime | 43611986           | 12              | 713                         | True                 | retained      | 1.0                      | False               | Protein kinase domain | PF00028      |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv   Protein Fusion: in frame {CCDC6:RET}  | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name        | partner_gene | frame_status         | target_role         | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                            | lost_domains                | disrupted_domains           | source_annotation_text                                                                                                                                     | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------------|--------------|----------------------|---------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|---------------------------------------------|-----------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-004<br>9588-T01-I<br>M6-159 | P-0049<br>588-T0<br>1-IM6 |            | RET-<br>CCD<br>C6  | CC<br>DC6    | out-<br>of-fr<br>ame | five<br>_pr<br>ime  | 436094<br>37       | 10              | 627                         | True                   | lost          | 0.0                      | False               | PF000<br>28                                 | Protein<br>kinase<br>domain |                             | RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-1522 2:CCDC6inv   Protein Fusion: out of frame {RET:CCDC6}                         | NA                           |
| EVT-P-001<br>5435-T01-I<br>M6-160 | P-0015<br>435-T0<br>1-IM6 |            | RET-<br>CCD<br>C6  | CC<br>DC6    | in-fr<br>ame         | three<br>_pr<br>ime | 436109<br>69       | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                 | PF0002<br>8                 |                             | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv   Protein fusion: in frame (CCDC6-RET) | NA                           |
| EVT-P-000<br>5556-T01-I<br>M5-161 | P-0005<br>556-T0<br>1-IM5 |            | RET-<br>CCD<br>C6  | CC<br>DC6    | in-fr<br>ame         | three<br>_pr<br>ime | 436118<br>28       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                 | PF0002<br>8                 |                             | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv   Protein fusion: in frame (CCDC6-RET) | NA                           |
| EVT-P-006<br>3502-T01-I<br>M7-162 | P-0063<br>502-T0<br>1-IM7 |            | RET-<br>CTN<br>NA1 | CTN<br>NA1   | unk<br>now<br>n      | five<br>_pr<br>ime  | 436220<br>38       | 19              | 1019                        | False                  | retained      | 1.0                      | False               | Protein<br>kinase<br>domain;<br>PF0002<br>8 |                             |                             | RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827::chr10:g.43622038)   Protein Fusion: mid-exon {RET:CTNNA1}             | NA                           |
| EVT-P-006<br>3802-T01-I<br>M7-163 | P-0063<br>802-T0<br>1-IM7 |            | RET-<br>CTN<br>NA3 | CTN<br>NA3   | out-<br>of-fr<br>ame | five<br>_pr<br>ime  | 436175<br>66       | 16              | 934                         | True                   | disrupted     | 0.7482269<br>5035461     | True                | PF000<br>28                                 |                             | Protein<br>kinase<br>domain | RET (NM_020975) - CTNNA3 (NM_013266) rearrangement: c.2801+102:RET_c.1375-342 98:CTNNA3inv   Protein Fusion: out of frame {RET:CTNNA3}                     | NA                           |
| EVT-P-003<br>0396-T02-I<br>M6-164 | P-0030<br>396-T0<br>2-IM6 |            | RET-<br>DOC<br>K1  | DO<br>CK1    | unk<br>now<br>n      | three<br>_pr<br>ime | 436083<br>20       | 9               | 556                         | False                  | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                 | PF0002<br>8                 |                             | DOCK1 (NM_001380) - RET (NM_020975)Rearrangement: c.2515+295:RET_c.1668:RETdup   Protein Fusion: mid-exon {DOCK1:RET}                                      | NA                           |
| EVT-P-001<br>3886-T01-I<br>M5-165 | P-0013<br>886-T0<br>1-IM5 |            | RET-<br>EML<br>4   | EML<br>4     | unk<br>now<br>n      | three<br>_pr<br>ime | 436100<br>67       | 11              | 673                         | False                  | retained      | 1.0                      | False               | Protein<br>kinase<br>domain                 | PF0002<br>8                 |                             | EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.43610067)   Protein fusion: mid-exon (EML4-RET)             | NA                           |
| EVT-P-001<br>4808-T03-I<br>M6-166 | P-0014<br>808-T0<br>3-IM6 |            | RET-<br>ERC<br>C6  | ERC<br>C6    | out-<br>of-fr<br>ame | five<br>_pr<br>ime  | 436112<br>16       | 12              | 713                         | True                   | lost          | 0.0                      | False               | PF000<br>28                                 | Protein<br>kinase<br>domain |                             | RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+682 2:ERCC6inv   Protein Fusion: out of frame {RET:ERCC6}                         | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name        | partner_gene   | frame_status         | target_role             | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                 | lost_domains                             | disrupted_domains | source_annotation_text                                                                                                                                | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------------|----------------|----------------------|-------------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|--------------------------------------------------|------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>4808-T02-I<br>M6-167 | P-0014<br>808-T0<br>2-IM6 |            | RET-<br>ERC<br>C6  | ERC<br>C6      | out-<br>of-fr<br>ame | five<br>_pr<br>ime      | 436112<br>16       | 12              | 713                         | True                   | lost          | 0.0                      | False               | PF000<br>28                                      | Prot<br>ein<br>kina<br>se d<br>oma<br>in |                   | RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+682 2:ERCC6inv   Protein Fusion: out of frame {RET:ERCC6}                    | NA                           |
| EVT-P-001<br>4808-T01-I<br>M6-168 | P-0014<br>808-T0<br>1-IM6 |            | RET-<br>ERC<br>C6  | ERC<br>C6      | out-<br>of-fr<br>ame | five<br>_pr<br>ime      | 436112<br>16       | 12              | 713                         | True                   | lost          | 0.0                      | False               | PF000<br>28                                      | Prot<br>ein<br>kina<br>se d<br>oma<br>in |                   | RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+682 2:ERCC6inv   Protein fusion: out of frame (RET-ERCC6)                    | NA                           |
| EVT-P-004<br>5809-T02-I<br>M6-169 | P-0045<br>809-T0<br>2-IM6 |            | RET-<br>FBXL<br>7  | FBX<br>L7      | in-fr<br>ame         | five<br>_pr<br>ime      | 436222<br>69       | 19              | 1063                        | True                   | retai<br>ned  | 1.0                      | False               | Protei<br>n<br>kinase<br>domai<br>n; PF0<br>0028 |                                          |                   | RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622269)   Protein Fusion: in frame {RET:FBXL7}           | NA                           |
| EVT-P-004<br>5809-T06-I<br>M6-170 | P-0045<br>809-T0<br>6-IM6 |            | RET-<br>FBXL<br>7  | FBX<br>L7      | in-fr<br>ame         | five<br>_pr<br>ime      | 436223<br>23       | 19              | 1063                        | True                   | retai<br>ned  | 1.0                      | False               | Protei<br>n<br>kinase<br>domai<br>n; PF0<br>0028 |                                          |                   | RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622323)   Protein Fusion: in frame {RET:FBXL7}           | NA                           |
| EVT-P-002<br>5215-T01-I<br>M6-171 | P-0025<br>215-T0<br>1-IM6 |            | RET-<br>GEMI<br>N5 | GE<br>MIN<br>5 | in-fr<br>ame         | thre<br>e<br>_pr<br>ime | 436110<br>67       | 11              | 712                         | True                   | retai<br>ned  | 1.0                      | False               | Protei<br>n<br>kinase<br>domai<br>n              | PF0<br>002<br>8                          |                   | GEMIN5 (NM_015465) - RET (NM_020975) rearrangement: t(5;10)(q33.2;q11.21)(chr5:g.154274993::chr10:g.43611067)   Protein Fusion: in frame {GEMIN5:RET} | NA                           |
| EVT-P-002<br>7417-T01-I<br>M6-173 | P-0027<br>417-T0<br>1-IM6 |            | RET-<br>KCN<br>MA1 | KCN<br>MA1     | out-<br>of-fr<br>ame | thre<br>e<br>_pr<br>ime | 436081<br>58       | 9               | 550                         | True                   | retai<br>ned  | 1.0                      | False               | Protei<br>n<br>kinase<br>domai<br>n              | PF0<br>002<br>8                          |                   | KCNMA1 (NM_001161352) - RET (NM_020975) rearrangement: c.378+15627: KCNMA1_c.1649-143:RET   Protein Fusion: out of frame {KCNMA1:RET}                 | NA                           |
| EVT-P-004<br>3704-T01-I<br>M6-174 | P-0043<br>704-T0<br>1-IM6 |            | RET-<br>KIF5<br>B  | KIF<br>5B      | in-fr<br>ame         | five<br>_pr<br>ime      | 436103<br>31       | 11              | 712                         | True                   | lost          | 0.0                      | False               | PF000<br>28                                      | Prot<br>ein<br>kina<br>se d<br>oma<br>in |                   | RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv   Protein Fusion: in frame {RET:KIF5B}                           | NA                           |
| EVT-P-000<br>6226-T02-I<br>M5-178 | P-0006<br>226-T0<br>2-IM5 |            | RET-<br>NCO<br>A4  | NC<br>OA4      | in-fr<br>ame         | thre<br>e<br>_pr<br>ime | 436118<br>93       | 12              | 713                         | True                   | retai<br>ned  | 1.0                      | False               | Protei<br>n<br>kinase<br>domai<br>n              | PF0<br>002<br>8                          |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139: RETdup   Protein fusion: in frame (NCOA4-RET)                             | NA                           |
| EVT-P-000<br>1151-T01-I<br>M3-179 | P-0001<br>151-T0<br>1-IM3 |            | RET-<br>NCO<br>A4  | NC<br>OA4      | in-fr<br>ame         | thre<br>e<br>_pr<br>ime | 436116<br>45       | 12              | 713                         | True                   | retai<br>ned  | 1.0                      | False               | Protei<br>n<br>kinase<br>domai<br>n              | PF0<br>002<br>8                          |                   | RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2137-387_c.1840-75dup   Protein fusion: in frame (NCOA4-RET)                                      | NA                           |

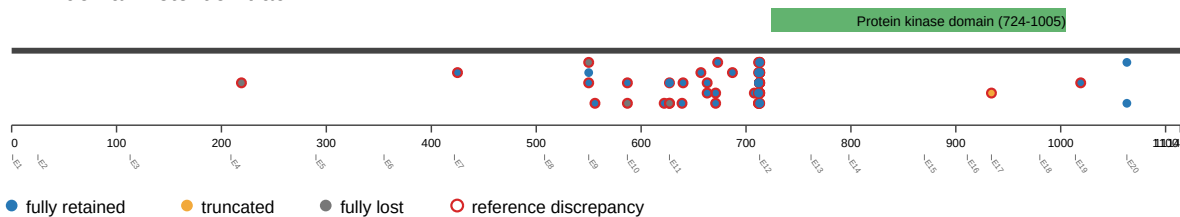
| event_id                   | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|----------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0009592-T01-I M5-180 | P-0009592-T01-IM5 |            | RET-NCOA4   | NCOA4        | unknown      | threep_rime | 43611143           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET exons 12=20: c.872:NCOA4_c.2136+621:RETdup   Protein fusion: mid-exon (NCOA4-RET)            | NA                           |
| EVT-P-0001155-T01-I M3-181 | P-0001155-T01-IM3 |            | RET-NCOA4   | NCOA4        | in-frame     | threep_rime | 43610941           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2136+757_c.715-617dup   Protein fusion: in frame (NCOA4-RET)                                                     | NA                           |
| EVT-P-0022818-T01-I M6-182 | P-0022818-T01-IM6 |            | RET-NCOA4   | NCOA4        | unknown      | threep_rime | 43611746           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused with RET exons 12-20): c.618+42: NCOA4_c.2137-860:RETdup   Protein Fusion: mid-exon {NCOA4:RET} | NA                           |
| EVT-P-0002145-T01-I M3-183 | P-0002145-T01-IM3 |            | RET-PDCD10  | PDCD10       | out-of-frame | five_p_rime | 43609394           | 10              | 627                         | True                   | lost          | 0.0                      | False               | PF00028               | Protein kinase domain |                   | RET (NM_020975) - PDCD10 (NM_145860) translocation: t(10;3)(q11.21;q26.1)(chr10:g.43609394::chr3:g.167419184)   Protein fusion: out of frame (RET-PDCD10)            | NA                           |
| EVT-P-0001898-T01-I M3-184 | P-0001898-T01-IM3 |            | RET-RASSF4  | RASSF4       | in-frame     | five_p_rime | 43611749           | 12              | 713                         | True                   | lost          | 0.0                      | False               | PF00028               | Protein kinase domain |                   | RET (NM_020975) - RASSF4 (NM_032023) deletion: c.2137-283_c.138+357del   Protein fusion: in frame (RET-RASSF4)                                                       | NA                           |
| EVT-P-0030396-T02-I M6-185 | P-0030396-T02-IM6 |            | RET-RBPMS   | RBPMS        | unknown      | threep_rime | 43610018           | 11              | 657                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | RBPMS (NM_001008712) - RET (NM_020975) Rearrangement : t(8;10)(p21.1,q11.21)(chr8:g.30412656::chr10:g.43610018)   Protein Fusion: mid-exon {RBPMS:RET}               | NA                           |
| EVT-P-0029222-T01-I M6-212 | P-0029222-T01-IM6 |            | RET-RUFY1   | RUFY1        | unknown      | threep_rime | 43610037           | 11              | 663                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | RUFY1 (NM_025158) - RET (NM_020975) Rearrangement : t(5;10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037)   Protein Fusion: mid-exon {RUFY1:RET}                 | NA                           |
| EVT-P-0015310-T02-I M6-213 | P-0015310-T02-IM6 |            | RET-RUFY3   | RUFY3        | out-of-frame | threep_rime | 43610427           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655316::chr10:g.43610427)   Protein fusion: out of frame (RUFY3-RET)            | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role      | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains                | disrupted_domains | source_annotation_text                                                                                                                                                                                  | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|-----------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>5310-T04-I<br>M6-214 | P-0015<br>310-T0<br>4-IM6 |            | RET-RUF<br>Y3   | RUF<br>Y3    | unknown      | threep<br>_prime | 436104<br>27       | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | RUFY3 (NM_001037442) -<br>RET (NM_020975)<br>rearrangement: t(4;10)(q13.3;<br>q11.21)(chr4:g.71655251::chr<br>10:g.43610427)   Protein<br>Fusion: mid-exon<br>{RUFY3:RET}                               | NA                           |
| EVT-P-002<br>5609-T01-I<br>M6-215 | P-0025<br>609-T0<br>1-IM6 |            | RET-SH<br>ROOM3 | SHR<br>OOM3  | in-frame     | threep<br>_prime | 436115<br>45       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | SHROOM3 (NM_020859) -<br>RET (NM_020975)<br>rearrangement: t(4;10)(q21.1;<br>q11.21)(chr4:g.77664882::chr<br>10:g.43611545)   Protein<br>Fusion: in frame<br>{SHROOM3:RET}                              | NA                           |
| EVT-P-005<br>3530-T01-I<br>M6-216 | P-0053<br>530-T0<br>1-IM6 |            | RET-SLC<br>4A4  | SLC<br>4A4   | out-of-frame | fivep<br>_prime  | 436104<br>39       | 11              | 712                         | True                   | lost          | 0.0                      | False               | PF000<br>28                 | Protein<br>kinase<br>domain |                   | RET (NM_020975) - SLC4A4<br>(NM_001134742)<br>rearrangement: t(4;10)(q13.3;<br>q11.21)(chr4:g.72412310::chr<br>10:g.43610439)   Protein<br>Fusion: out of frame<br>{RET:SLC4A4}                         | NA                           |
| EVT-P-000<br>2145-T01-I<br>M3-217 | P-0002<br>145-T0<br>1-IM3 |            | RET-TFG         | TFG          | in-frame     | threep<br>_prime | 436094<br>37       | 10              | 627                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | RET (NM_020975) - TFG<br>(NM_001195478)<br>translocation: t(10;3)(q11.21;q<br>12.2)(chr10:g.43609437::chr3:<br>g.100450751)   Protein fusion:<br>in frame (TFG-RET)                                     | NA                           |
| EVT-P-002<br>1980-T01-I<br>M6-218 | P-0021<br>980-T0<br>1-IM6 |            | RET-TIM<br>M23B | TIM<br>M23B  | out-of-frame | threep<br>_prime | 436105<br>50       | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | NCOA4 (NM_001145260) -<br>RET (NM_020975) fusion<br>(NCOA4 exons 1-8 fused to<br>RET exons 12- 20): c.763-591<br>:NCOA4_c.2136+366:RETdup<br>  Protein Fusion: out of frame<br>{TIMM23B:RET}            | NA                           |
| EVT-P-001<br>7716-T01-I<br>M6-219 | P-0017<br>716-T0<br>1-IM6 |            | RET-TIM<br>M23B | TIM<br>M23B  | out-of-frame | threep<br>_prime | 436111<br>72       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | NCOA4 (NM_001145260) -<br>RET (NM_020975) fusion<br>(NCOA4 exons 1-7 fused with<br>RET exons 12-20): c.618+42:<br>NCOA4_c.2137-860:RETdup  <br>Protein Fusion: out of frame<br>{TIMM23B:RET}            | NA                           |
| EVT-P-002<br>5531-T01-I<br>M6-220 | P-0025<br>531-T0<br>1-IM6 |            | RET-TIM<br>M23B | TIM<br>M23B  | out-of-frame | threep<br>_prime | 436113<br>75       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | NCOA4 (NM_001145260)<br>RET (NM_020975) fusion: c.6<br>19-108NCOA4_c.2137-657RE<br>Tdup   Protein Fusion: out of<br>frame {TIMM23B:RET}                                                                 | NA                           |
| EVT-P-003<br>4704-T01-I<br>M6-222 | P-0034<br>704-T0<br>1-IM6 |            | RET-TRIM<br>24  | TRI<br>M24   | in-frame     | threep<br>_prime | 436118<br>27       | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain | PF0<br>002<br>8             |                   | TRIM24 (NM_015905) - RET<br>(NM_020975) Fusion (TRIM24<br>exon 9 fused with RET<br>exon12) : t(7;10)(q34;q11.21)(<br>chr7:g.138243011::chr10:g.43<br>611827)   Protein Fusion: in<br>frame {TRIM24:RET} | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name         | partner_gene    | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                                                  | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|---------------------|-----------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-004<br>6790-T01-I<br>M6-223 | P-0046<br>790-T0<br>1-IM6 |            | RET-<br>UXS<br>1    | UXS<br>1        | unknown      | five_prime  | 43600430           | 4               | 219                         | False                  | lost          | 0.0                      | False               |                       | Protein kinase domain | PF00028           | RET (NM_020975) - UXS1 (NM_001253875) rearrangement: t(2;10)(q12.2;q11.21)(chr2:g.106728260::chr10:g.43600430)   Protein Fusion: mid-exon {RET:UXS1}                                                    | NA                            |
| EVT-P-002<br>9222-T03-I<br>M6-224 | P-0029<br>222-T0<br>3-IM6 |            | RUF<br>Y1-R<br>ET   | RUF<br>Y1       | unknown      | three_prime | 43610037           | 11              | 663                         | False                  | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | RUFY1 (NM_025158) - RET (NM_020975) fusion: t(5;10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037)   Protein Fusion: mid-exon {RUFY1:RET}                                                            | NA                            |
| EVT-P-003<br>1340-T02-I<br>M6-225 | P-0031<br>340-T0<br>2-IM6 |            | RUF<br>Y2-R<br>ET   | RUF<br>Y2       | in-frame     | three_prime | 43611559           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | RUFY2 (NM_017987) - RET (NM_020975) fusion: c.1045-640:RUFY2_c.2137-473:RETinv   Protein Fusion: in frame {RUFY2:RET}                                                                                   | NA                            |
| EVT-P-002<br>6614-T01-I<br>M6-227 | P-0026<br>614-T0<br>1-IM6 |            | SPE<br>CC1L<br>-RET | SPE<br>CC1<br>L | in-frame     | three_prime | 43611575           | 12              | 713                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(10q11.21;22q11.23)(chr10:g.43611575::chr22:g.24740413)   Protein Fusion: in frame {SPECC1L:RET} | NA                            |
| EVT-P-003<br>6400-T06-I<br>M6-228 | P-0036<br>400-T0<br>6-IM6 |            | SPE<br>CC1L<br>-RET | SPE<br>CC1<br>L | in-frame     | three_prime | 43610353           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | SPECC1L (NM_001145468) - RET (NM_020975) fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056)   Protein Fusion: in frame {SPECC1L:RET}                                                   | NA                            |
| EVT-P-003<br>6400-T04-I<br>M6-229 | P-0036<br>400-T0<br>4-IM6 |            | SPE<br>CC1L<br>-RET | SPE<br>CC1<br>L | in-frame     | three_prime | 43610353           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056)   Protein Fusion: in frame {SPECC1L:RET}      | NA                            |
| EVT-P-003<br>6400-T03-I<br>M6-230 | P-0036<br>400-T0<br>3-IM6 |            | SPE<br>CC1L<br>-RET | SPE<br>CC1<br>L | in-frame     | three_prime | 43610353           | 11              | 712                         | True                   | retained      | 1.0                      | False               | Protein kinase domain | PF00028               |                   | SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056)   Protein Fusion: in frame {SPECC1L:RET}      | NA                            |

## domain retention outliers

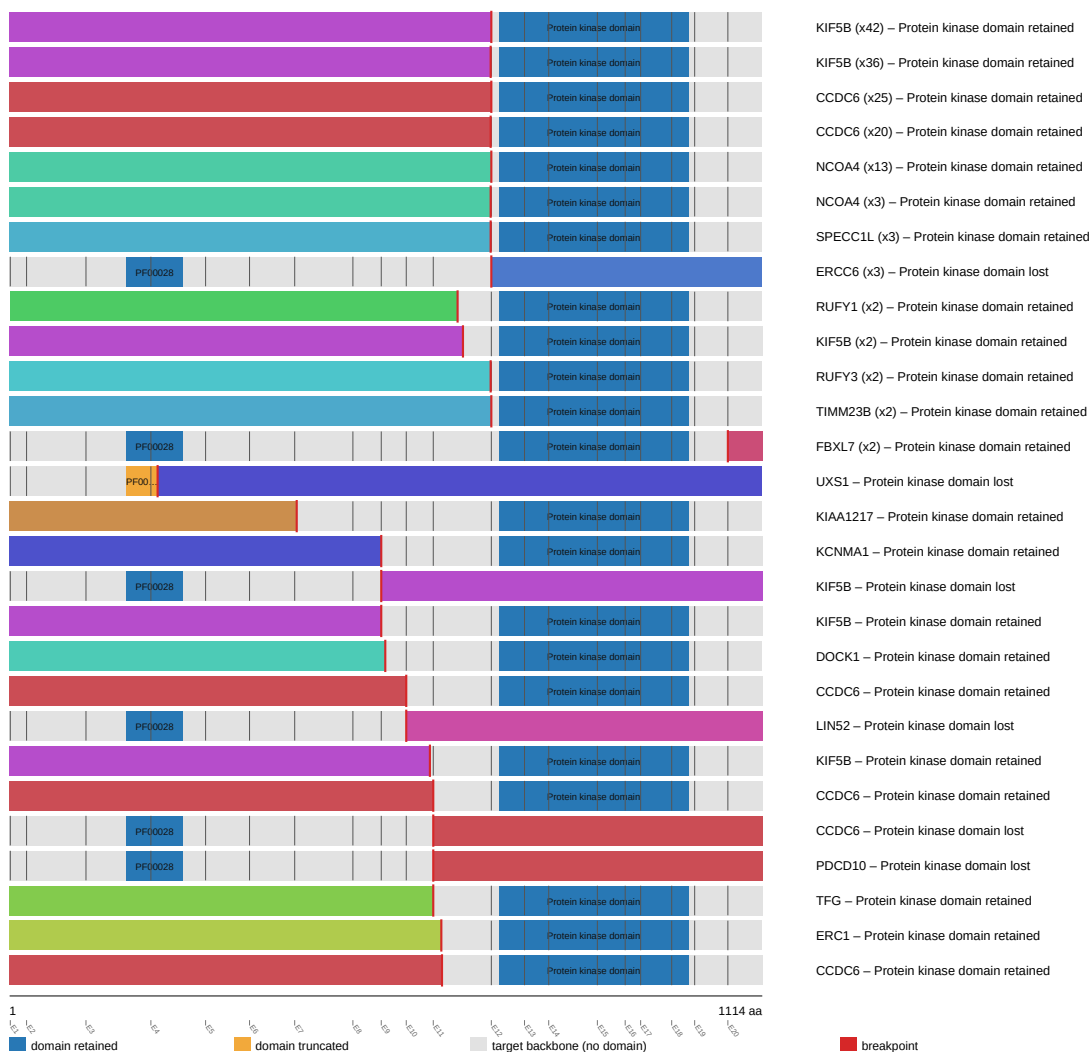
RET domain-retention track



## fusion schematic

RET fusion-transcript schematic

partner block → breakpoint → retained RET portion (domain-colored, exon ticks)



Showing the top 28 of 52 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk\_impact\_50k\_2026

